

APELEVEL / ENGINEERING & SECURITY

WHISPER MARKET × RAGE REALMZ

EXECUTIVE ENGINEERING REPORT

Security foundation • Custom Amoy deployment • tTILT airdrop

28 OCTOBER 2026 | 358 TRACKED DEVELOPMENT MILESTONES

Executive summary

Overview. ApeLevel completed a security-first development progression for Whisper Market and RAGE REALMZ, including 63 reported passing Whisper automated tests, a purpose-built Polygon Amoy deployment portal, deployment of a capped tTILT contract, and a 24-recipient holder airdrop through Coinbase Wallet, and the public blockchain transaction proof for that distribution. ApeLevel used a dedicated testnet deployment portal separate from the RAGE REALMZ game portal, then retired the deployment interface after the work was completed.

WORKSTREAM	DOCUMENTED RESULT
Whisper automated testing	63 passes: 50 Node.js Oracle-related + 13 Foundry Solidity
Custom deployment portal	16 recorded passing HTTP/config/header assertions across overlapping test runs, plus supplementary functional checks
tTILT token	Deployed on Polygon Amoy (chain 80002), cap of 500,000
Airdrop allocation	24 NFT-owner wallets; 494,129.555122971848455571 tTILT
Historical mainnet cutoff	Polygon block 96,321,652
Deployment transaction	Status 1 (success) confirmed by a receipt read
Airdrop proof	Public Amoy Polygonscan transaction reference; completion confirmed by project owner

Evidence standard and scope

This report distinguishes public on-chain references, terminal/browser results recorded during development, and unverified completion claims. The deployment receipt was returned through a Polygon Amoy RPC with status 1. The airdrop transaction has a public Amoy Polygonscan proof link, and project owner confirmation establishes the reported completion of the distribution. This report links that transaction as the primary evidence, while a separate individual-wallet reconciliation has not been independently repeated by the report author.

The 358-step numbering represents tracked engineering milestones. Detailed terminal logs are present for the later implementation stages, while the earliest stages are summarized in project context. This report does not invent one-to-one descriptions for undocumented step numbers.

Public proof index

- Whisper repository
- tTILT contract & balances
- Successful deployment transaction
- Airdrop batch transaction
- RAGE QUITTERS NFT collection
- Polygon mainnet TILT

01 Why ApeLevel prioritized security

The Whisper Market combines off-chain game resources (Water, Wood, inventories and production history) with blockchain-based token settlement. ApeLevel did not want game resources independently tradable outside the intended economy; TILT and tTILT supply the on-chain component. This division creates an Oracle trust boundary: neither an internal resource adjustment nor a wallet message alone should be enough to authorize final settlement.

- Treasury round-trip arbitrage: a market that buys Water back for more than it sells it could be drained through repeated purchase-and-resale cycles. Planned defenses included spreads, bounded pricing, inventory limits and validated trades.
- Manipulated listings: artificially low player asks should not automatically dictate treasury reference prices; executed trades and anomalous-input checks matter.
- Failure atomicity: inventory reservation, token confirmation and settlement must be coordinated so interrupted operations do not strand or duplicate assets.

02 Whisper Oracle and treasury safeguards

The developing Oracle foundation uses persistent SQLite-related state, explicit transaction transitions, idempotent processing, proof representations, Ed25519 signatures and network-specific verification. The intended final-settlement rule requires sanitized proof publication to the official Whisper repository followed by independent verification. If publication fails or is rate-limited, settlement must remain nonfinal rather than bypassing the gate.

The treasury foundation validates deployment-network configuration, requires a deployed token address, separates administrator and guardian roles, and establishes pause/unpause permissions. These are initial safeguards; the full live-value deposit, withdrawal and settlement surface was not yet implemented.

Whisper test result — distinct from portal checks

TEST GROUP	REPORTED PASSES
Oracle and supporting Node.js tests	50
Foundry Solidity tests	13
Whisper automated total	63

The 63 are reported passes for Whisper only. They support controlled Amoy experimentation; they are not a third-party security audit, nor evidence that all future settlement functions have been tested. Complete individual test names and assertions were not available in the material supplied for this report.

03 Custom Amoy portal — built separately

ApeLevel built tools/amoy-deployer as a standalone CommonJS HTTP service, deliberately separate from the existing RAGE REALMZ game portal. A restricted-access Codespaces deployment portal was used. The portal served only explicit allowlisted files and blocked direct requests to sensitive paths. Its browser controller used injected EIP-1193 wallet requests; private-key entry was not part of the portal workflow.

Recorded deployment-portal checks

TEST RUN	PASSING ASSERTIONS	EVIDENCE
Static file access	5	/, /app.js, /style.css = 200; /.env, /server.js = 404
Security headers	4	Root HTTP 200; X-Frame-Options DENY; allowed Amoy RPC; frame-ancestors none
Configuration access	3	/config.json HTTP 200; chain/network configuration valid; .env unavailable
Later endpoint access	4	/, /config.json, /app.js, /test-tilt.json = HTTP 200
TOTAL ASSERTIONS	16	Includes overlapping checks; not 16 separate audits

Supplementary successful checks included Node JavaScript syntax, bytecode/ABI integrity, wallet-address and chain restrictions, verified configuration, and blockchain reads. These are not folded into an invented larger numeric audit total.

Portal failures and ApeLevel resolutions

- Missing artifact (HTTP 404): the running process did not serve the newly allowlisted /test-tilt.json route. An updated server restart restored the endpoint.
- Port conflict (EADDRINUSE): an earlier instance of the deployment service still occupied the selected network interface. ApeLevel stopped the stale listener without interfering with the game portal.
- Network switching: the original wallet controller rejected non-Amoy chains but did not prompt a switch. wallet_switchEthereumChain and add-chain fallback were added.
- Coinbase Wallet CSP blocks: walletlink WebSocket and Polygon drpc requests were refused. Specific endpoints were allowed and header delivery rechecked. Permitting inline wallet styles is an explicit security tradeoff.
- Browser fetch failures: local HTTP endpoint testing and browser console messages isolated wallet/forwarded-site issues instead of prompting another contract build.

04 tTILT smart-contract build and deployment

The initial ERC-20 test token used OpenZeppelin ERC20 and AccessControl. It allowed role-restricted minting but no cap. ApeLevel introduced MAX_SUPPLY = 500,000 × 10^18 and checked minting against remaining supply. A batchMint(address[],uint256[]) function, capped at 100 recipients per batch, was added for distribution.

Toolchain friction included a missing forge PATH entry and a nonexistent temporary Solidity compiler at /tmp/apelevel-solc-0.8.38. Foundry 1.8.6 built the Solidity 0.8.38 code successfully after the path was overridden. A compiler warning for an uninitialized total was addressed; remaining require-in-loop lint warnings were not build failures.

ARTIFACT MILESTONE	CREATION BYTECODE
Earlier uncapped token	3,710 bytes
Capped token	3,867 bytes
Capped and batch-mint-enabled token	4,734 bytes

Gas deficiency overcome

A deployment attempt failed with insufficient funds. The Amoy wallet held 0.012226449713157221 test POL, versus an estimated deployment cost of 0.027227200148702 4 test POL (1,047,200 estimated gas). ApeLevel obtained additional faucet funds and retried the same capped build.

On-chain deployment evidence

FIELD	OBSERVED VALUE
Amoy contract	0xc1dd560375459b19d6437f076f1708e286bb5d67
Deployment block	51,365,293
Transaction status	1 — success
Gas consumed	1,037,796
Deployment signer	0x486735d3defd63f4759f1d88df4e11e2ec4698eb
Postdeployment total supply	0 tTILT
MAX_SUPPLY getter	500,000 tTILT
Signer holds MINTER_ROLE	True

The verified contract deployment did not mint tokens. The first actual supply increase was reserved for a separate airdrop transaction. Open deployment proof

05 Snapshot, RPC restrictions and redesign

The deployment-time cutoff was tied to Amoy confirmation time, not to the numeric Amoy block height. The corresponding last Polygon mainnet block identified at or before the timestamp was 96,321,652. The mainnet timestamp was one second before the recorded deployment time.

The first holder-discovery method reconstructed token balances from Transfer logs, starting at the original mint block 67,745,563. The original event minted 500,000 TILT to the development wallet. A 500-block scanner maintained a durable checkpoint and reconciled the partial running supply, but 28,576,090 blocks implied approximately 57,153 requests.

- PublicNode: historical state unavailable; later requests explicitly demanded an archive-access personal token.

- Pocket Network: returned the original mint event and initially accepted a 100,000-block query, but later ranges failed with “block range exceeds maximum 50000” and “block range exceeds configured limit.”
- Preserved progress: a checkpoint recorded 361,500 blocks, 34 transfer events, 24 tracked addresses and 500,000 reconstructed TILT, but this was only 1.265% of the required range.
- Decision: stop the expensive global scan and use the already configured Ethereum RAGE QUITTERS collection to identify a smaller test cohort.

06 NFT-owner cohort and 1:1 allocation

The configured Ethereum collection address was 0x70a95FbCe719d9B8C02DBe40C71fdDCc333a4e80. The collection reported 44 tokens. ownerOf(1–44) returned owners for all queried IDs, deduplicating to 28 wallets. A historical Polygon balanceOf check for NFT #1 owner returned 25,000 TILT at snapshot block 96,321,652.

All 28 candidate NFT-owner wallets were queried at that fixed Polygon block. Four had zero TILT, leaving 24 positive-balance recipients. The resulting allocation was 494,129.555122971848455571 tTILT, with 5,870.444877028151544429 remaining outside the NFT-owner cohort.

ALLOCATION DETAIL	RESULT
NFTs examined	44
Unique NFT owner addresses	28
Positive-balance recipient wallets	24
Calculated distribution	494,129.555122971848455571 tTILT
Undistributed cap allowance	5,870.444877028151544429 tTILT
Development-wallet allocation	334,000.000220318930375931 tTILT
Source block	Polygon mainnet 96,321,652

Eligibility caveat: this is an NFT-owner-based cohort, not an exhaustive list of every Polygon TILT holder. ApeLevel must not claim the entire 500,000-token mainnet holder population was replicated by this distribution.

Local allocation integrity

The saved allocation file was checked for chain IDs, correct target token, positive amounts, distinct addresses, exact raw-unit total and cap compliance. Recorded SHA-256: 67cc6533f92524fef9093c702efec231cfd2e77515127b91297363daeba0d2ba. This proves integrity relative to the saved file, not independent correctness of every upstream ownership query.

07 Airdrop UI and Coinbase Wallet transaction

The original Deploy button was retired to avoid unnecessarily redeploying tTILT. ApeLevel added a separate airdrop panel displaying the deployed contract, historical snapshot, exact prepared total and wallet count. A display controller kept signing disabled until allocation data loaded. A separate signing controller initially was not connected, explaining repeated locked-button reports.

The batchMint transaction was ABI-encoded with ethers using the full integer amounts. The browser check initially rejected the transaction because the prepared JSON lacked the allocationSha256 field expected by the signer. The file was corrected and every encoded recipient and amount was cross-checked against the allocation JSON.

The custom portal displayed the airdrop transaction hash after Coinbase Wallet approval. The project owner subsequently confirmed successful completion, and ApeLevel retained the public Amoy Polygonscan transaction proof for the distribution.

Public airdrop reference

View airdrop transaction on Amoy Polygonscan

Transaction hash: 0xd9f435644075ddeaa8b0c3abc1937a4aa0e540db279157532576ac7ac9719759

Public proof: the linked Amoy Polygonscan transaction identifies the batch-minting operation and is the public reference supplied for the successful airdrop. The project owner confirmed completion. As a further audit enhancement, ApeLevel can archive the transaction receipt, supply change and recipient-by-recipient Transfer reconciliation; that supplementary review was not performed independently for this report.

08 Engineering decisions that mattered

- Do not expose deployment controls in the game: use a dedicated testnet deployment portal separate from the game portal, then shut it down when finished.
- Do not embed private keys: sign through a connected Coinbase Wallet and honor user transaction approval.
- Do not mint before determining eligibility: the token launched with zero supply and a separate distribution path.
- Do not confuse test and production chains: Amoy 80002 for token deployment; Polygon 137 for historical balances; Ethereum 1 for NFT ownership.
- Do not round transaction amounts: show six decimals for readability but encode all 18 decimals exactly.
- Do not mistake a cap for deduplication: a 500,000 cap prevents oversupply but cannot alone prevent duplicate allocation attempts within remaining capacity.
- Do not blindly trust RPC providers: archive permissions and log-range limits differ by provider, block era and request.
- Do not present 63 Whisper tests as a complete audit: they validate tested components; the deployed portal added separate checks and also has untested boundaries.

09 Remaining engineering priorities

- P0 — Archive distribution proof: preserve the publicly linked airdrop transaction and capture the receipt, recipient transfer events, totalSupply and individual balances in a formal reconciliation record.
- P0 — Return to the game portal: keep the separate deployment portal shut down and continue RAGE REALMZ development in its existing game environment; preserve deployment scripts without re-enabling signing.
- P1 — Show Amoy tTILT in the game: read-only balance display, approved contract address, correct decimals and explicit separation from mainnet TILT.
- P1 — Protect wallet/NFT workflows: preserve Ethereum read-only NFT ownership checks and Polygon testnet token transactions without confusing chain prompts.
- P1 — Implement Whisper trade lifecycle: synchronized off-chain resource reservations, payment proof checks, replay resistance and recoverable settlement states.
- P1 — Enforce public-proof authorization gate: real published proof and independent verification before final settlement, including GitHub outage and RPC failure cases.
- P2 — Treasury and Oracle integration tests: test network binding, permissions, pause, accounting and realistic adversarial transaction sequences.
- P2 — Economic simulation: reproduce and eliminate buyback round-trip arbitrage and manipulated-listing behavior.
- P2 — Independent review before production: assess wallet exposure, CSP tradeoffs, game authentication, Oracle signatures, treasury administration and mainnet settlement.

Final assessment

What ApeLevel established: a 63-pass Whisper test foundation, a custom-built isolated Amoy portal with 16 recorded HTTP/config/security-header assertions, a confirmed capped testnet contract deployment, historical NFT-cohort allocation data and a successful holder airdrop reported by the project owner with a public Polygonscan transaction proof.

What remains: archive recipient-level airdrop reconciliation, integrate the test token into the RAGE REALMZ game portal, finish the Whisper transaction lifecycle and complete deeper security review before real-value use.

The engineering significance lies in the deliberate sequence: economic threat modeling, verifiable Oracle design, automated component testing, restricted testnet infrastructure, artifact and wallet verification, historical allocation, controlled transaction signing, and retirement of deployment tooling. Testnet progression was justified by targeted evidence, not a claim that production risks had disappeared.

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